Packet Sniffing

Aim:

To implement a packet sniffer using Raw Socket and display basic IP packet information.

Procedure.

1. Create a Raw Socket using socket().
2. Bind it to the local IP address.
3. Enable packet capture using SIO_RCVALL.
4. Receive packet using recv()
5. Extract source IP, destination IP, TTL, Protocol, and Packet Size.
6. Display the Capture Packet information
7. Stop using Ctrol + C.

Code:

```
#include <stdio.h>
#include <Winsock2.h>
#include <ws2tcpip.h>
#pragma <ws2tcpip.h>
#pragma comment (lib,"ws2-32.lib").
```

```
int main ()
{
  WSADATA wsa;
  SOCKET s;
  Char buffer [65536]
  DWORD flag = 1;
  WSAStartup (MAKEWORD (2,2), &wsa);
  S= socket (AF_INET; SOCK_RAW, IPPROTO_IP);
  if (s== INVALID_SOCKET).
  {
   Printf ("Socket Creation failed : %d \n", WSAGetLastError());
   return 1;
  }
  Char host [256];
  get host name (host, size of (host));
  Struct hostent * h = get host by name (host);
  Struct sockaddr_in addr;
  addr.sin_family = AF_INET;
  addr.sin_addr = * (Struct in_addr *) h -> h_addr;
  bind (s, (struct sockaddr*) &addr, size of (addr));
  WSAIoctl (s, SIO_RCVALL, & flag, size of (flag);
            NULL, 0, & flag, NULL, NULL);
```

output

Ether / IP / UDP 172.16.79.235.44021 > 239.0.0.8:5

Source IP : 172.16.79.235

Destination IP : 239.0.0.8

Protocol : 17

Packet length : 1078

Ether / IP / UDP 172.16.79.235:44021 > 239.0.0.8.5172

Soura IP : 172.16.79.235

Destination IP : 239.0.0.8

Protocol : 17

Packet length : 1078

Ether / IP / UDP 172.16.79.235:51276 > 239.0.0.8:5172

source IP : 172.16.79.235

Destination IP : 239.0.0.8

Protocol : 17

Packet length : 1078

```
Printf("RAW Socket Packet Sniffer started \n");
while(1)
    {
    int size = recv.(s, buffer, sizeof(buffer),0);
if (size >0)
  {
    Printf("\n Packet bize : %d bytes \n", size);
    Printf ("Packet capture successfully. \n");
  }
}
(loresocket(s);
WSA cleanup();
return 0;
}.
```

Result:

Thus, a RAW socket packet sniffer was successfully implemented and network packets were captured successfully.